

Apuntes – Semana 11

Ejemplo 1:

Determine si el vector

$$(-1, 6, 1) = \vec{v}$$

pertenece al espacio generado por $\vec{v}_1$ $\vec{v}_2$ $\vec{v}_3$

$$S = \{(1, -1, 2), (3, -1, 8), (2, -1, 5)\}$$

$$a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3 = \vec{v} \rightarrow \text{s.e.}$$

Sea $a, b, c \in \mathbb{R}$ tal que:

$$a(1, -1, 2) + b(3, -1, 8) + c(2, -1, 5) = (-1, 6, 1)$$

$$(a, -a, 2a) + (3b, -b, 8b) + (2c, -c, 5c) = (-1, 6, 1)$$

$$\Rightarrow \begin{cases} a + 3b + 2c = -1 \\ -a - b - c = 6 \\ 2a + 3b + 5c = 1 \end{cases} \Rightarrow \text{No soluciones}$$

$$(-1, 6, 1) \notin \text{LIN}(S)$$

Tiene Sol.
 $\rightarrow \vec{v} \in \text{LIN}(S)$
 $\rightarrow$ No hay Sol.
" Inconsistente
 $\Rightarrow \vec{v} \notin \text{LIN}(S)$

Ejemplo 2:
Determine si

$$\{(2, 3, 1), (1, -1, 2), (7, 3, 8)\}$$

son LD o LI.

Fórmula: $a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3 = \vec{0}$

$\hookrightarrow$ S.E. $\rightarrow$ Sol Trivial $(0, 0, 0)$ LI

$\hookrightarrow$ Sol no trivial LD

Sea $a, b, c \in \mathbb{R}$ tal que:

$$a(2, 3, 1) + b(1, -1, 2) + c(7, 3, 8) = (0, 0, 0)$$

$$\begin{cases} 2a + b + 7c = 0 \\ 3a - b + 3c = 0 \\ a + 2b + 8c = 0 \end{cases} \Rightarrow \text{Inf. Sol (no trivial)}$$

$\therefore$ L.D.

Ejemplo 3:

Determinar si

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ -2 & 0 \end{pmatrix} \right\}$$

es LD o LI.

$$a \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + b \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} a & 3b \\ -2c & -a \end{pmatrix}_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{2 \times 2}$$

$$\left\{ \begin{array}{c|c} a=0 & 3b=0 \\ \hline -2c=0 & -a=0 \end{array} \right\}$$

$$\begin{array}{l} a=0 \checkmark \\ -a=0 \\ a=0 \\ -1 \\ a=0 \checkmark \end{array}$$

$$\begin{array}{l} 3b=0 \\ b=\frac{0}{3} \\ b=0 \checkmark \\ \hline -2c=0 \\ c=0 \\ -2 \\ c=0 \end{array}$$

$$\therefore \text{Sol trivial } a=b=c=0 \\ (0,0,0)$$

$\Rightarrow$ L.I.

Ejemplo 4:

Determinar si

$$p(x) = x^2 - 3x + 2, \quad q(x) = x^2 - 2x, \quad r(x) = x^2 - x$$

es LD o LI

Sea $a, b, c \in \mathbb{R}$ tal que:

$$a \cdot p(x) + b \cdot q(x) + c \cdot r(x) = 0$$

$$a(x^2 - 3x + 2) + b(x^2 - 2x) + c(x^2 - x) = 0$$

$$ax^2 - 3ax + 2a + bx^2 - 2bx + cx^2 - cx = 0$$

$$x^2(a + b + c) + x(-3a - 2b - c) + (2a) = 0x^2 + 0x + 0$$

$$\begin{cases} a + b + c = 0 \\ -3a - 2b - c = 0 \\ 2a = 0 \end{cases}$$

$$a = \frac{0}{2} \Rightarrow a = 0$$

$$\begin{cases} b + c = 0 \\ -2b - c = 0 \end{cases} \Rightarrow \begin{matrix} b = 0 \\ c = 0 \end{matrix}$$

$\therefore$ Solución Trivial $a = b = c = 0$

$\Rightarrow$ L.I. $\Rightarrow$ BASE

Ejemplo 5:

Determine si

$$\{(1, 2, 3), (2, 4, 6), (0, 1, 1)\}$$

son LD o LI.

Sea $a, b, c \in \mathbb{R}$ /

$$a(1, 2, 3) + b(2, 4, 6) + c(0, 1, 1) = (0, 0, 0)$$

$$\begin{cases} a + 2b = 0 \\ 2a + 4b + c = 0 \\ 3a + b + c = 0 \end{cases} \Rightarrow \text{Infinitas Sol} \\ \Rightarrow \text{No es trivial} \\ \Rightarrow \text{L.D.}$$

Ejemplo 6:

Determine si

$$S = \{(1, 1, 0), (0, 1, 1), (1, 2, 1)\}$$

forma una base de $\mathbb{R}^3$.

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 2, 1) = (0, 0, 0)$$

$$\begin{cases} a + 0b + c = 0 \\ a + b + 2c = 0 \\ 0a + b + c = 0 \end{cases}$$

$\Rightarrow$ Inf. Sol
no trivial

$\Rightarrow$ L.D. no es un BASE

Ejemplo 7:

Determine los valores de t para que

$$v_1 = (0, t-1, -1), \quad v_2 = (1, -1, t-2), \quad v_3 = (t, -1, -1)$$

sean linealmente independientes.

En $\mathbb{R}^3$; tres vectores son LI $\Leftrightarrow \det(A) \neq 0$

$$A = \begin{pmatrix} 0 & 1 & t \\ t-1 & -1 & -1 \\ -1 & t-2 & -1 \end{pmatrix}_{3 \times 3}$$

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 0 & 1 & t \\ t-1 & -1 & -1 \\ -1 & t-2 & -1 \end{vmatrix} \neq 0$$

$(+)a_{11} \quad (-)a_{12} \quad (+)a_{13} \rightarrow a_{ij} \rightarrow (-1)^{i+j}$

$$\begin{aligned} (-1)^{1+1} &= 1 \\ (-1)^{1+2} &= -1 \\ (-1)^{1+3} &= 1 \end{aligned}$$

$$\Rightarrow 0 \cdot \begin{vmatrix} -1 & -1 \\ t-2 & -1 \end{vmatrix} - 1 \cdot \begin{vmatrix} t-1 & -1 \\ -1 & -1 \end{vmatrix} + t \cdot \begin{vmatrix} t-1 & -1 \\ -1 & t-2 \end{vmatrix} \neq 0$$

$$\begin{aligned} & -1 \cdot [-1 \cdot (t-1) - (-1) \cdot (-1)] + t \cdot [(t-1)(t-2) - (-1) \cdot (-1)] \neq 0 \\ & -1 \cdot [-t+1-1] + t \cdot [t^2-2t-t+2-1] \neq 0 \end{aligned}$$

$$-1 \cdot [-] \quad t \cdot [t^2 - 3t + 1] \neq 0$$

$$t + t^3 - 3t^2 + t \neq 0$$

$$t^3 - 3t^2 + 2t \neq 0$$

$$a = 1$$

$$b = -3$$

$$c = +2$$

$$d = 0$$

$$t_1 \neq 0$$

$$t_2 \neq 1$$

$$t_3 \neq 2$$

$$t \in \mathbb{R} - \{0, 1, 2\}$$